CMS-0042-NC **June 16, 2025**

Solicitation of Public Comments

Centers for Medicare & Medicaid Services (CMS), Assistant Secretary for Technology Policy (ASTP) Office of the National Coordinator for Health Information Technology (ONC)

Health Technology Ecosystem

Submitted To:

https://www.regulations.gov

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This request for information (RFI) seeks input from the public regarding the market of digital health products for Medicare beneficiaries as well as the state of data interoperability and broader health technology infrastructure. All comments received before the close of the comment period are available for viewing by the public, including any personally identifiable or confidential business information that is included in a comment.

GDIT is pleased to respond to this solicitation of public comments to provide the Centers for Medicare & Medicaid Services feedback about the Assistant Secretary for Technology Policy (ASTP) Office of the National Coordinator for Health Information Technology (ONC) Health Technology Ecosystem.

B. Patients and Caregivers

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

a. **What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?**

The Department of Health & Human Services (HHS) has made significant investments in modernizing secure, virtual access to some of the largest health services research data repositories in the country, notably the Centers for Medicare & Medicaid Services (CMS) Chronic Conditions Warehouse and Virtual Research Data Center (CCW VRDC). This comprehensive environment houses over 400 billion rows of Medicare and Medicaid claims data dating back to 1999, with more than one billion new records added monthly. The warehouse integrates more than 40 data sources, including beneficiary enrollment, Medicare and Medicaid administrative claims, prescription drug events, managed care encounter data, and assessment data from nursing home and home health care settings. It also includes supplemental sources such as the National Death Index and social risk indicators.

The environment supports advanced analytic capabilities, including Snowflake, SAS, R, and Python through a container-based Jupyter platform, as well as Virtual Desktop Infrastructure and data visualization tools like Databricks and Tableau. These tools allow researchers and developers to conduct meaningful real-world research with actionable evidence. Additionally, the CMS Blue Button on FHIR (BBonFHIR) initiative, supported by a Data-as-a-Service (DaaS) platform, enables beneficiaries to connect their individual Medicare Part A, B, and D claims data with trusted applications. This DaaS infrastructure is designed to allow data providers to publish analytics-ready data products without requiring subscribers to build data processing pipelines.

b. **What are specific sources, other than claims and clinical data, that would be of highest value, and why?**

c. **What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?**

Many types of data that are highly valuable to patients, caregivers, and application developers remain difficult to access. Clinical data, such as encounter notes, operative reports, and real-time

appointment schedules, are often siloed in proprietary systems or buried in unstructured formats. Moreover, even when available, this data is not always interoperable or timely enough to drive personalized care or meaningful research.

A particularly high-value but underutilized category of data is **Health-Related Social Needs (HRSN)** data. This includes metrics such as housing instability, food insecurity, transportation challenges, and utility access—factors that directly impact health outcomes but are rarely captured in traditional claims or clinical systems. By systematically collecting and integrating HRSN data, organizations can:

- Inform whole-person care by addressing both clinical and social determinants of health
- Predict health outcomes and costs through early identification of risk factors
- Support value-based care models
- Enable equitable resource allocation
- Foster collaboration between healthcare and social service providers

Standardized tools such as the **AHC HRSN Screening Tool** (developed by CMS), **PRAPARE** (Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences), and the **Joint Commission HRSN Screening Question Bank** provide frameworks for consistent data collection across organizations and geographies.

To meaningfully advance data accessibility, interoperability, and integration, HHS has the opportunity to build upon its existing investments in the CCW VRDC and DaaS platforms by:

- Expanding integration with electronic health information, environmental datasets, and social determinants of health data;
- Streamlining and automating data access across both clinical and community-based data systems;
- Supporting API-based data sharing models to accelerate innovation in care delivery and patient empowerment.

These steps would enable a more holistic and actionable evidence base, bridging the gap between clinical insight and real-world conditions that affect patient health.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

There are several CMS data sources that would significantly enhance the Blue Button 2.0 API's value for building digital products for patients and caregivers:

- **Medicare Part C (Medicare Advantage) Claims:** Currently, Blue Button 2.0 does not provide Part C claims data, which covers a substantial and growing portion of Medicare beneficiaries. Including this would offer a more complete view of patient care and utilization.
- **Clinical Data including Electronic Health Record (EHR) Data:** Access to clinical data, such as lab results, vital signs, immunizations, and care plans, beyond claims, would enable richer, more actionable insights for both patients and care teams.

- **Prior Authorization and Administrative Data:** Adding data related to prior authorizations, referrals, and eligibility checks would streamline administrative workflows and improve care coordination.
- **Social Determinants of Health (SDOH):** Information about social and environmental factors impacting health—such as housing, food security, and transportation—could help developers create tools that address the broader context of patient health.
- **Provider Directories:** Up-to-date information on provider networks, specialties, and availability would help patients make more informed decisions when selecting providers.
- **Estimated Cost of Care:** Including estimated cost information based on historical claims data and/or CMS fee schedules could allow patients to anticipate out-of-pocket costs and make more cost-conscious healthcare decisions.
- **Utilization Summary Data:** Sharing aggregated, de-identified utilization metrics, such as frequency of specific services by provider and geographic location, would give patients valuable context about the typical patterns of care in their area and by specific providers, helping them better understand and navigate their care options.

Together, these data enhancements would significantly improve the utility of the Blue Button 2.0 API, empowering patients and caregivers with more complete, understandable, and actionable information to support decision-making, care planning, and health management.

a. **What difficulties are there in accessing or utilizing these data sources today?**

Developers and stakeholders face several challenges in accessing and utilizing these data sources:

- **Fragmentation and Incomplete Data:** The absence of Part C data and limited clinical information means the API does not provide a full patient record, reducing its utility for comprehensive care management.
- **Workflow Integration:** Surfacing data at the right time within provider and patient workflows remains a challenge.
- **Data Quality and Timeliness:** Claims data can be delayed and may not reflect real-time clinical events, limiting its usefulness for immediate decision-making.

b. **What suggestions do you have to improve the Blue Button 2.0 API experience?**

Suggestions to Improve the Blue Button 2.0 API Experience:

- **Expand Data Scope:** Incorporate Medicare Part C claims, clinical data from EHRs, and administrative datasets, such as prior authorizations and eligibility checks.
- **Enhance Real-Time Data Access:** Work toward reducing claims data lag and, where possible, provide more real-time clinical data feeds.
- **Improve Documentation and Developer Support:** Offer more comprehensive guides, code samples, and sandbox environments to lower the technical barrier for new developers.

- **Strengthen Security and Consent Management:** Continue to improve OAuth 2.0 protocols and user consent flows to ensure privacy and compliance while making access easier for authorized apps.
- **Facilitate Workflow Integration:** Develop tools and resources to help integrate API data directly into EHRs, patient portals, and care coordination platforms, supporting both patient and clinician use cases.

c. **Is there non-CMS data that should be included in the API?**

Non-CMS data that should be included:

- **State Medicaid Data:** Further leverage existing Medicaid claims and coverage information would benefit dual-eligible populations and provide a more complete view of healthcare utilization.
- **Private Insurance Data:** Where feasible, support for importing data from commercial insurers would help patients with coverage transitions or multiple sources of insurance.
- **Patient-Generated Health Data:** Incorporating data from personal health devices (e.g., wearables, home monitoring) could enrich the patient record and support personalized care.
- **Community and Social Services Data:** Linking resources, such as transportation, housing, and nutrition support, could help address social determinants of health.

By addressing these areas, the Blue Button 2.0 API can become a more powerful tool for developers, enabling the creation of digital products that are more comprehensive and timely.

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

- **What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?**

Bulk FHIR data exports from electronic health records (EHRs) offer a compelling opportunity to reduce the burden on clinicians participating in CMS quality reporting programs. By enabling continuous and standardized data exchange, this approach could streamline the reporting process, eliminating the need for clinicians to manually generate and submit custom reports. This automation aligns with CMS's goals of reducing administrative overhead and allowing clinicians to focus more on patient care.

However, the implementation of Bulk FHIR data exports is not without its challenges. Ensuring the accuracy and completeness of quality measure calculations from exported data is paramount. EHRs—even those from the same vendor—can vary significantly across sites due to local customizations, and clinician documentation practices differ widely. While structured data supports automated reporting, it often fails to capture the full richness of the patient narrative. If

clinicians must double-document to preserve this narrative while also meeting structured data requirements, the intended burden reduction may be undermined.

Another potential downside lies in the cost of enabling and maintaining these data exports. Depending on EHR vendor pricing models, clinicians or healthcare organizations may face new or increased costs, even if other more labor-intensive reporting processes are phased out.

Importantly, the broader quality measurement ecosystem could also be affected. Qualified Clinical Data Registries (QCDRs) currently play a key role in developing and maintaining specialty-specific quality measures, often at a faster pace than CMS's Measures Under Consideration (MUC) process. QCDRs rely heavily on revenue from clinician participation, which could decline if simplified FHIR-based reporting leads clinicians to disengage. This could severely limit the pipeline of new quality measures, particularly for underrepresented specialties.

The CMS Merit-based Incentive Payment System (MIPS) measure inventory is dynamic, and measures frequently become "topped out," meaning there's little room for additional improvement. As such, an ongoing mechanism for new measure development is essential. While CMS has expressed an intent to reduce the overall number of measures, the current MUC process alone may not be agile enough to sustain a responsive and relevant measure set.

In summary, Bulk FHIR data exports hold promise for easing clinician reporting burdens, but careful attention must be paid to data integrity, documentation practices, cost implications, and the preservation of the quality measure development ecosystem—particularly the role of QCDRs—in order to ensure the long-term success and equity of CMS quality programs.

PA-4. What would be the value to payers of a nationwide provider directory that included FHIR end points and used digital identity credentials?

The primary benefits to payers of a nationwide provider directory would be reduced administrative burden related to the maintenance and upkeep of their provider directories and improved accuracy of the provider data within. The use of FHIR to exchange data supports CMS's Office of Health Experience and Interoperability's mission to promote interoperability to support informed decision making by patients, caregivers, providers and a more efficient health care system overall. While FHIR provides for a secure exchange of data, the use of digital identity credentials would provide an additional layer of security.

C. Technology Vendors, Data Providers, and Networks

1. Ecosystem

TD-2. Regarding CMS Data, to stimulate developer interest—

a. **What additional data would be most valuable if made available through CMS APIs?**

As mentioned under section PC-8, HHS has made significant investments in modernizing secure, virtual access to some of the largest health services research data repositories in the country, namely the CMS CCW VRDC. This comprehensive data warehouse contains 100% of Medicare and Medicaid claims data dating back to 1999 (more than 400 billion rows of data). One billion rows of data are added to the CCW VRDC monthly, which has storage capacity of over 12

petabytes (PBs) of AWS Cloud provisioned storage. More than 40 data sources are integrated within the warehouse, including beneficiary enrollment/eligibility, Medicare and Medicaid administrative claims, prescription drug events, managed care encounter data, nursing home and home health assessment data, as well as supplemental reference code sources, National Death Index data, and a full library of research data products.

FHIR and cloud native secure file transfer tools currently available through the CMS Portal could be leveraged to collect quality measures, clinical data, survey data, and social determinants of health data from other CMS systems or sources from other entities via cloud sharing or APIs for integration with electronic health records. Leveraging this existing HHS investment and expanding the integration of new data sources (e.g., clinical quality measures, electronic health information, environmental, and related sources of social determinants of health) would provide an opportunity to increase accessibility and establish interoperability and integration of clinical data to support more real-world research and actionable evidence. Ultimately, this rich data source could be leveraged for real-time access to beneficiary utilization of services, electronic health record, and benefit information through a digital health product.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A nationwide provider directory of FHIR endpoints would allow payers to more efficiently maintain current provider directories for the benefit of consumers who need important information when selecting plans or providers within a plan's network. Data, such as the provider's name, specialty, address, phone number, availability, and accepted insurance, are important for consumers to make informed choices. CMS could make such a resource available based on provider data already available in National Plan and Provider Enumeration System (NPPES). The use of FHIR and clearly defined processes can ensure continual accuracy.

A nationwide FHIR endpoints directory would serve as critical infrastructure for healthcare interoperability by:

- **Improving Access & Understanding:** A centralized directory would eliminate the current fragmented discovery process where providers must individually negotiate data sharing agreements. This creates a standardized catalog enabling automated discovery of data sources, reducing integration timelines from months to weeks. For claims data, specifically, it would provide clear mapping between clinical care delivery points and billing entities, resolving the common disconnect between where care occurs and how it's reported financially.

  The directory would enable real-time patient data aggregation across multiple providers during care episodes, particularly valuable for emergency scenarios and care transitions. It also supports population health analytics by providing a complete view of data source coverage and gaps.

- **Streamlining Payer Operations:** A nationwide FHIR endpoints directory would significantly streamline payer operations by enabling automated provider directory updates. Currently, payers manually maintain separate directories through surveys and

phone verification - a resource-intensive process prone to data staleness. With standardized FHIR endpoints, payers could automatically pull updated provider information, including specialty changes, new locations, and network participation status. This ensures consumers have accurate, real-time data when selecting plans or providers within networks, directly impacting healthcare access decisions.

Leveraging existing National Plan and Provider Enumeration System (NPPES) data as a foundation, CMS could establish authoritative provider demographic information while FHIR endpoints provide the technical connectivity layer. This dual approach ensures both data accuracy and technical interoperability.